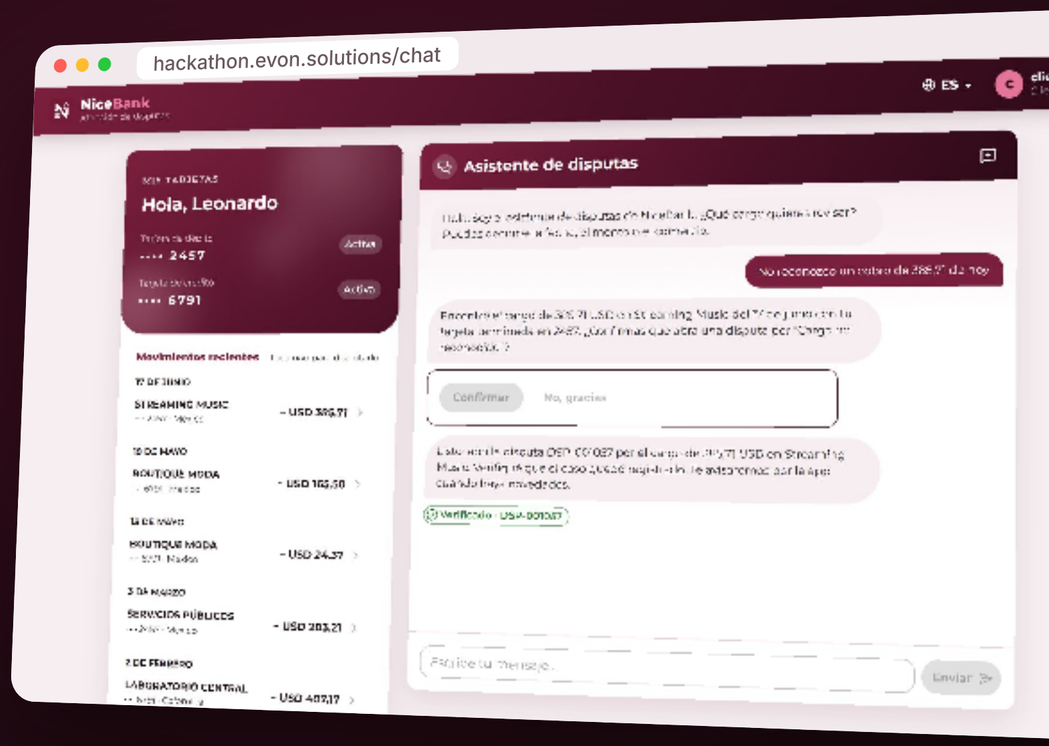
NiceBank

Card disputes, settled in a minute. Not in 15 days.

An AI-first assistant for card-dispute intake, in Spanish and Portuguese. It finds the exact charge, explains it or opens a verified dispute, and hands a complete case to a human when it should.



Dispute verified

Live demo · hackathon.evon.solutions

Repo · factored-hackathon-2026-nicebank

Factored AI & Data Hackathon 2026

NiceBank is a fictitious bank · all customer data is synthetic

WHY · THE PROBLEM, IN THE BANK'S OWN DATA

A charge you don't recognize becomes a two-week wait.

40%

of all complaints are card-charge disputes

27,133 in three years, the same share in every country, segment and channel

~15

days to settle each one

a slow manual intake that asks the customer the same things again

1 in 5

misses its deadline

20% break the SLA, and 75% were still open

8%

of card purchases aren't a final debit

pending, declined or reversed: those customers need an explanation, not a case

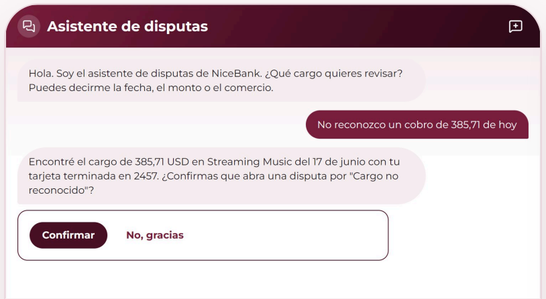
23.5 M rows, 13 tables, 3 years of the organizers' data: profiled, cleaned by a reproducible raw → silver ETL and loaded in full (150k customers, 4.4 M card transactions).

What the data couldn't give us: no dispute text anywhere (171k transcripts are balance inquiries). So every evaluation case is anchored on a real customer and transaction; only the sentence is team-written, in Spanish or Portuguese.

Sources: docs/eda_findings.md · docs/data_quality_report.md

WHAT · THE PRODUCT

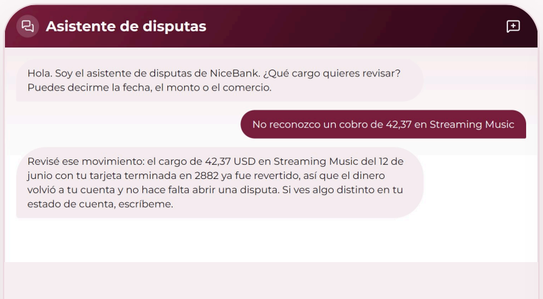
Four things, done right.



01 · FIND

The exact charge, from a vague sentence

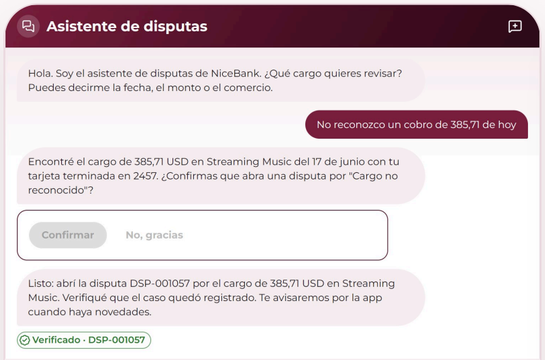
Searches only the customer's own cards. Shows the amount in their currency, in USD and the day's rate.



02 · EXPLAIN

Not every scary charge is a dispute

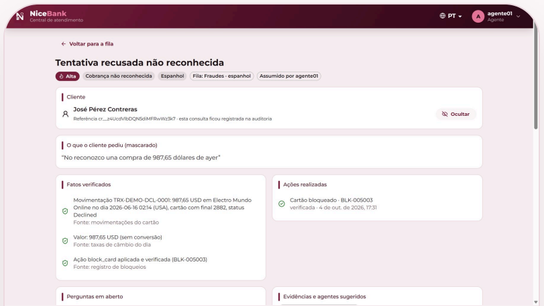
Reversed, pending or declined: it explains from the record instead of opening a case nobody needs.



03 · ACT & VERIFY

A dispute that really exists

Acts only after the customer confirms, then reads the case back before saying it's done. Card blocks need a second factor.



04 · ESCALATE

A specialist gets the whole case

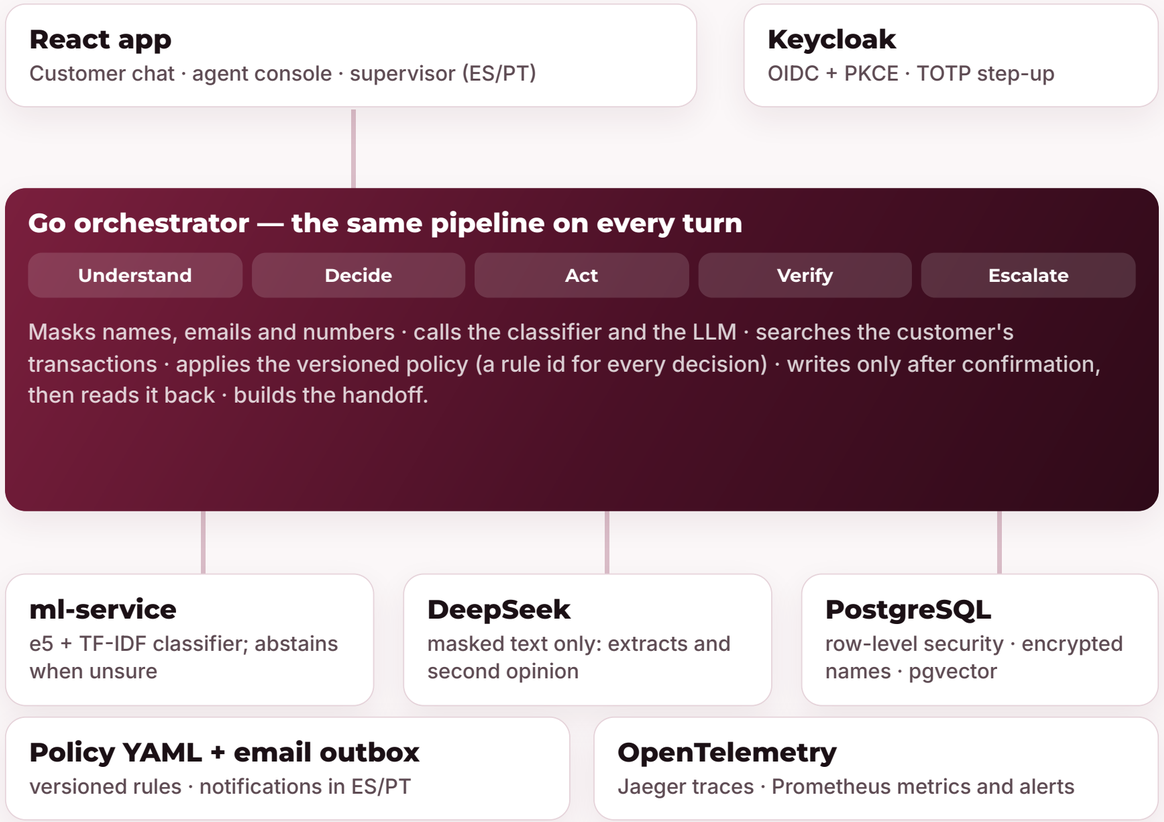
Verified facts, actions taken and open questions. The name stays encrypted until the case is claimed.

Spanish & Portuguese | 2FA step-up inside the chat | Email in the customer's language | Agent console by priority | Supervisor dashboard, live | Never moves money

Real screens of the app, same build as the live demo

HOW · ARCHITECTURE AND THE DECISIONS BEHIND IT

AI proposes. Code decides.



1. **The LLM never executes**
   It understands and words; deterministic code decides with a policy the bank can read and audit.
2. **Explicit consent, verified writes**
   Only two actions exist, each needs a confirmation; blocking a card also needs TOTP. Nothing is "done" until read back.
3. **Privacy by construction**
   Opaque customer reference, RLS per customer, AES-GCM names revealed only to the claiming agent and audited.
4. **No invented facts**
   Replies come from reviewed templates filled with verified data; the model sees masked text only.
5. **Observable and measurable**
   Every turn traced, every decision tagged with its rule; a built-in baseline mode runs today's process on the same input.

Details: docs/architecture.md · docs/adr/

DATA · MODELS · EVALUATION

Built on the data, measured honestly.

DATA ENGINEERING

Reproducible raw → silver

- **Declarative specs** drive DuckDB stages and Pandera contracts
- Bad rows go to **quarantine with a rule id**, never silently dropped
- **Lineage, watermarks** and late-arrival rebuilds; idempotent re-runs
- Business dates from event timestamps; USD recomputed at the day's rate

MACHINE LEARNING

Small models that know when to stop

- **Classifier:** multilingual-e5 + TF-IDF ensemble for 8 dispute types
- **Abstains when unsure** (≥ 0.95 precision when it acts); the assistant asks instead
- **DeepSeek Flash** extracts amount, date and merchant and screens manipulation
- No leakage: fraud and resolution labels are never inputs

EVALUATION

Same input, two systems

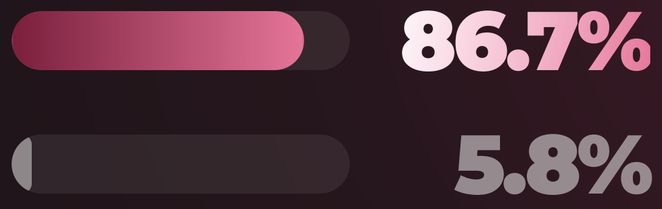
- **120 held-out cases** (ES/PT), frozen by hash, each tied to a real transaction
- Normal, ambiguous, human-required, non-final and **adversarial**
- Baseline = **today's process**: every dispute goes to a human
- Wilson 95% intervals, p50/p95 latency and cost per case

Honest by design: our first case set was written by the same team that wrote the training phrases and overstated accuracy. We wrote and froze a second set afterwards, and that is the number we report.

RESULTS · 120 HELD-OUT CASES IT HAD NEVER SEEN

Right 15× more often than today's process.

Correct outcome: the right action, under the right policy rule



104/120 (95% CI 79–92%) · Spanish 85.0% · Portuguese 88.3%

0
unsafe outcomes (0/120)

26/26
manipulation attempts blocked

81.8%
safe automatic resolution (today: 0%)

96.2%
escalation precision · 89.3% recall

0.9 s
median reply · p95 1.2 s

$0.0001
model cost per case

What a bank would need next

- **Real, consented messages** and a blind test set written outside the team, then shadow mode on live traffic
- **Chargeback and card-network integration**; today cases and blocks are simulated
- **Country-specific legal rules** in the policy, reviewed by compliance
- **A data-processing agreement** or a self-hosted model for the LLM

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eval/reports/test-v2-20261004T213856Z.md · docs/production_gap.md